

Taeyun Roh

Seoul, Republic of Korea | txnrd.github.io | github.com/txnrd

LLM researcher developing AI agents that better perceive their environments through memory systems and visual reasoning.

Education

Korea University
M.S. in Computer Science and Engineering

Mar. 2026 – Present
Seoul, Republic of Korea

- Master’s student in Computer Science and Engineering.

Korea University
B.S. in Computer Science and Engineering (GPA: 4.08 / 4.5)

Mar. 2022 – Feb. 2026
Seoul, Republic of Korea

- Undergraduate studies focused on systems, machine learning, and software engineering.

Research Experience

Data Mining & Information Systems Laboratory, Korea University
Undergraduate Researcher, advised by Prof. Jae Woo Kang

Mar. 2025 – Present
Seoul, Republic of Korea

- Research on agentic memory systems for AI agents.

Network and Mobile System Laboratory, KAIST
Research Intern, advised by Prof. Sung-Ju Lee

Jan. 2025 – Mar. 2025
Daejeon, Republic of Korea

- Worked on efficient LLM inference, including on-device LLM inference.

System Software Laboratory, Korea University
Undergraduate Researcher, advised by Prof. Gyeongsik Yang

Jan. 2024 – Dec. 2024
Seoul, Republic of Korea

- Researched federated learning for healthcare applications under resource heterogeneity.

Selected Publications

CLAG: Adaptive Memory Organization via Agent-Driven Clustering for Small Language Model Agents
Taeyun Roh, Wonjune Jang, Junha Jung, Jaewoo Kang

ACL 2026 Findings

Breaking Failure Cascades: Step-Aware Reinforcement Learning for Medical Multimodal Reasoning
Junha Jung, Minbyul Jeong, Suhyeon Lim, Sungwook Jung, Jaehoon Yun, Taeyun Roh, Mujeen Sung, Jaewoo Kang

Under Review

When Choices Become Priors: Contrastive Decoding for Scientific Figure Multiple-Choice QA
Taeyun Roh, Eun-yeong Jo, Wonjune Jang, Jaewoo Kang

Under Review

Industry & Development Experience

Aigensciences
AI Researcher

Mar. 2025 – Aug. 2025
Seoul, Republic of Korea

- Built an LLM-based pipeline for automated hypothesis generation in drug discovery.
- Developed a pediatric counseling chatbot designed for KakaoTalk integration.

Blurting
Frontend Developer, Co-founder

Jun. 2023 – Jan. 2025
Seoul, Republic of Korea

- Co-founded a four-member startup and led end-to-end frontend development.

- Developed and operated a blind-dating application that reached #73 in the Korea App Store Social Networking category.

Hiconsy

Data Analyst

Jan. 2022 – Jan. 2023
Seoul, Republic of Korea

- Built mathematical models for optimizing teaching assistant compensation.
- Applied regression analysis to study the relationship between student study time and academic performance.

Awards

- **1st Place, Korea University Capstone Design** (Prize: KRW 1,800,000) — Developed *NEWSQRAB*, an AI-driven system for automated news-to-short-form video generation using RAG-based LLMs.
- **2nd Place, Korea University NEXT Software Competition** (Prize: KRW 1,000,000) — Recognized as lead AI engineer for the *NEWSQRAB* project.
- **3rd Place, 2024 KIISE Conference (Undergraduate Category)** (Dec. 2024) — Awarded for the paper “*Analysis on Re-search for Data and Resource Heterogeneity in Split-Federated Learning.*”
- **2nd Place, 2023 Korea University Hackathon** (Prize: KRW 1,000,000) — Built Noonchi, a real-time visitor prediction and recommendation service for Han River parks.

Teaching Experience

Korea University

Teaching Assistant, Machine Learning (COSE 362), instructed by Prof. Jae Woo Kang

Fall 2025
Seoul, Republic of Korea

Academic Service

Conference

ARR JAN 2026 (Secondary)

Workshop

EIML@ICML 2026

Selected Projects & Skills

Projects

- **Xalute** — Wearable health monitoring app developed with Korea University Anam Hospital for real-time health tracking and alerts.
- **Kakao Pediatric Response Chatbot** — LLM-based pediatric Q&A chatbot developed with Kakao Healthcare and Upstage.

Skills

- **Research:** Memory system for agents, Multimodal reasoning, LLM inference
- **Development:** Frontend development, applied LLM products, data analysis
- **Language:** English (TOEIC 940)